INDEX

1.- OBJECT AND SCOPE

2.- LEVELS OF DEVELOPMENT

2.1. 2D PROJECTS

2.2. 3D PROJECTS

3.- CHOICE OF LEVELS OF DEVELOPMENT

4.- SCOPE OF DOCUMENTS CAUSED ACCORDING TO ITS LEVEL.

4.1. PIPING DESIGN

4.2. PIPING MATERIALS

4.3. STRESS & SUPPORTS

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1.- OBJECT AND SCOPE

This Job Instruction has the purpose of defining levels of development of the Engineering services that the Piping Departments has to render, to prepare the documents that makes up the scope of the project of the following specialties:

- Piping Design
- Piping Materials
- Piping Stress and Supports

Its application is extended to all works of the project, excluding audits, reports and technical assistance.

This Job Instruction is part of the Intecsa Industrial Quality System and it has been prepared taking into account the ISO-9001 Code.

2.- LEVELS OF DEVELOPMENT

Depending on the level of detail required in the development of engineering services and the tool, following levels have been set:

2.1. Projects in 2D

- Level 1:

  Projects with a minimum detail level, defining a basic level of data and key elements of the installation, and will allow further development of the detailed engineering and may require provision of information. It applies to conceptual projects or basic engineering.

- Level 2:

  Projects with a medium detail level, with preliminary definition of equipments and main lines. The engineering may be developed in full during the detailed engineering phase. Applies to standard projects FEL / FEED.

- Level 3:

  Projects with sufficient detail to enable the construction and assembly of the installation without the need of any additional engineering services.

2.2. Projects in 3D

- Level 1

  Projects with a minimum detail level, defining a basic level of data and key elements of the installation.

  The elements modeled, must be valid for further development of detailed engineering level 2 and / or 3.

- Level 2:

  Projects with a medium level of detail.

  Modeled and design of all major equipment and lines. The engineering may be developed in full during the detailed engineering phase. Applies to standard projects FEL / FEED.

- Level 3:

  Projects with sufficient detail to enable the construction and assembly of the installation.

  Design and modeled of the elements, is made so that the extracted data model (drawings, isometrics, etc.) will be valid for the construction, without the need of additional engineering.

3.- CHOICE OF THE LEVEL OF DEVELOPMENT

Defining the level of development due to the management, according to the kind of project hired. (See PR-10).

4.- SCOPE OF THE DOCUMENTS PRODUCED ACCORDING TO THE LEVEL

The following describes the scope of detail required in the preparation of the various documents produced, depending on the level of development indicated.

4.1. Piping Design

DOCUMENT	Level 1	Level 2	Level 3
Plot Plan (Items: 3623 / 3630)	Orientation of plant and definition of the origin of coordinates, plot and battery boundaries situation and/or access location.	Ditto to Level1.	Ditto to Level 1.
	Disposition of interior streets, buildings and major structures indicating coordinates or minimum distances that constrain the final implant.	Disposition of all interior streets, and all buildings and structures, with position co-ordinates axes and identifying pillars.	Ditto to Level 2.
	Implant may be performed by BLOCKS.	Ditto to Level 1 with heights definition of structures and equipment.	Ditto to Level 2 and definition of access, disassembly and assembly areas, etc.
	Indication of special paving areas and canals or ditches that constrain the overall layout.	Ditto to Level 1 with localization of curbs, bunds, pits, etc.	Ditto to Level 2.
	Definition of pipe paths.	Ditto with definition of Tier 1 PIPE RACKS / PIPE TRACKS (widths and heights required).	Ditto to Level 2.
	Arrangement of main equipment and, if necessary, with their trading platforms and / or basic maintenance for the process, specifying minimum distances coordinates or conditioning the final implementation.	Representation and position coordinates of all facility equipment platform indicating major forms and heights.	Ditto to Level 2.
	General Notes and references basic installation.	List of equipment, general notes and references. Location of stations and eyewash showers.	Ditto to Level 2.
Key Plan (Item 3631)	Is determined depending on the design requirement (not necessary for projects of small volume).	Limits representing piping plan drawings and numbering thereof, to be performed on copy of the key plane.	Ditto to Level 2.

2D PROJECTS			
DOCUMENT	Level 1	Level 2	Level 3
Piping Plan View Drawings (Items 3624/3650)	Representation of the basic design of main or critical lines as required.	Ditto to Level 1, including the representation of equipment platforms, battery limits and major structures.	Ditto to Level 2, including all lines and elements shown in the P & ID's, as well as whole structures.
Piping studies for model (Item 3651)	No model will be made in projects of this level, except block model if it is required.	Simplified representation, reduced to the essential for interpretation of axes and/or outlines of buildings, structures, equipment, operating platforms and access roads. Schematic representation of the piping networks within the scope of the study, with indication and identification of only the main elements online (valves, filters, instruments, etc.), without defining constructive details.	Ditto to Level 2. Ditto to Level 2.
Rack Drawings (Item 3626 /3652)	Representation of main axes of gantry structures. Schematic representation of mains process and service pipes.	Ditto to Level 1, extending the representation of the operating plat-forms as appropriate (ex. battery limits). Representation of all pipes included in engineering diagrams, specifying and with identification of all components (valves, instruments, etc.). As a particular case, details and vertical elevation will be done for the correct definition of the battery limits.	Ditto to Level 2. Ditto to Level 2.

2D PROJECTS			
DOCUMENT	Level 1	Level 2	Level 3
Auxiliary Drawings, Vents and Drains (Item 3653)	It will not be made for projects with this level, except when some of these details may be imposed by certain special conditions of the process.	All those details that could be typified will be schematically defined, such as the installation of vents, drains and instrument connections, and it will be employed the standard drawings corresponding to the steam tracing circuits, cooling and sealing of pumps and agitators where appropriate.	Ditto to Level 2.
Isometrics (Items 3660 / 3661)	It will not be made for projects with this level (except critical lines if it is necessary).	It will be made only for major pipe diameters, as defined in the Coordination Procedure of the Project, and provided that the alternative of constructional details on plot plans has not been chosen.	It will be made for all pipes, as defined in the Coordination Procedure of the Project.

3D PROJECTS			
DOCUMENT	Level 1	Level 2	Level 3
Piping Plot Plans (Item 3650)	Obtaining piping drawings at this level will be conditional as designed in 3D.	Ditto to Level 1.	Ditto to Level 1.
Isometrics (Item 3660/ 3661)	Isometrics will not be edited at this level.	Isometrics will not be edited at this level (except critical lines if it is necessary).	Obtaining isometric piping at this level will be conditioned as designed in 3D.
3D Piping Design (Item 3640)	Basic design of main or critical pipes as required. Main equipments will be modeled.	Ditto to Level 1, extending the design to design platforms, battery limits, structures, etc. in its entirety. (It will take into account the space required by other specialties, such as cable trenches, underground pipes, FWS, etc.).	Ditto to Level 2 including all lines and elements represented in the P&ID's. The design of ditches, structures, trays, hardness boxes, etc. will be executed by the relevant specialty.
Rack Drawing (Item 3652)	Drawings will not be edited at this level.	Obtaining piping drawings at this level will be conditional on the designed in 3D, as required by Project.	Ditto to Level 1. Floor and elevation drawings will be edited.
Piping Measurements (Material Take-Off) (Item 3665)	See document BOMs, Item 3420.	Ditto to Level 1.	Ditto to Level 1.
Physical Model Assembly (Item 3670)	No model will be made in projects of this level. (except block model if required).	There will be mounting the model as defined in the IT-TU-17, Design and Construction of Models, revised for the project, if applicable.	Ditto to Level 2

3D PROJECTS			
DOCUMENT	Level 1	Level 2	Level 3
Information for other Departments (Item 3680)	As defined for each project. Information for other departments is included in the 3D model and will be obtained by each department directly from the model. No charged hours on this Item.	Ditto to Level 1.	Ditto to Level 1.

4.2 Piping Materials

DOCUMENT	Level 1	Level 2	Level 3
Material Specifications (Item 3410)	Defining classes of basic specifications for plant process, with general description of the main elements that compose them.	Complete definition of all kinds of material specifications, with full description of the elements that compose them.	Ditto to Level 2
Material Lists (Item 3420)	Making one MTO, with obtaining lists of materials including estimated amounts based on the level of Engineering and scope of pipe networks represented in engineering diagrams.	Performing 1 M.T.O. with obtaining lists of materials and performing of BM's for evaluation.	Performing 3 MTO's with obtaining complete lists of materials and performing R.M 's for purchase.
Items H (Item 3430)			Performing RM's of Items H.

4.3 Stress and supports

DOCUMENT	Level 1	Level 2	Level 3
Stress (Item 3510)	Calculations of analysis flexibility will be made only on the main lines of process that condition on laying of equipments defined in Plot Plan.	All necessary calculations of analysis flexibility will be to ensure the adequacy of the design of pipe networks.	Ditto to Level 2.
Supports (Item 3520)			On the plans will include all supports, anchors, guides or other structural design support for networks of pipes of 3 "(DN 80) and larger.
List Of Supports (Item 3530)			Support List will be made.
Special Elements (Item 3540)			Special elements shall be indicated in drawings derived from analysis flexibility calculations and its flexibility characteristics.